

A Technical Introduction to Attention is All You Need

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Section 1

"Attention Is All You Need" is a 2017 research paper in machine learning authored by eight scientists and engineers working at Google. The paper introduced a new deep learning architecture known as the transformer, based on the attention mechanism proposed in 2014 by Bahdanau et al. The transformer approach it describes has become the main architecture of a wide variety of artificial intelligence, including large language models. At the time, the focus of the research was on improving Seq2seq techniques for machine translation, but the authors go further in the paper, foreseeing the technique's potential for other tasks like question answering and what is now known as multimodal generative AI. Some early examples that the team tried their Transformer architecture on included English-to-German translation, generating Wikipedia articles on "The Transformer", and parsing. These convinced the team that the Transformer is a general-purpose language model, and not just good for translation. As of 2025, the paper has been cited more than 173,000 times, placing it among the top ten most-cited papers of the 21st century. After the paper was published by Google, each of the authors left the company to join other companies or to found startups.

Section 2

wherein p o s $\{\text{displaystyle } \rm \{pos\}\}$, i $\{\text{displaystyle } i\}$, d m o d e l $\{\text{displaystyle } d_{\rm \{model\}}\}$ correspond to the position of the word, the current dimension index, and the dimension of the model, respectively. The sine function is used for even indices of the embedding while the cosine function is used for odd indices. The resultant P E $\{\text{displaystyle } PE\}$ embedding is then added to the word at that corresponding position with respect to the current context window. The paper specifically comments on why this method was chosen describing: "We chose the sinusoidal version because it may allow the model to extrapolate to sequence lengths longer than the ones encountered during training."

Section 3

Multi-head attention In the self-attention mechanism, queries (Q), keys (K), and values (V) are dynamically generated for each input sequence (typically limited by the size of the context window), allowing the model to focus on different parts of the input sequence at different steps.

Multi-head attention enhances this process by introducing multiple parallel attention heads. Each attention head learns different linear projections of the Q, K, and V matrices. This allows the model to capture different aspects of the relationships between words in the sequence simultaneously, rather than focusing on a single aspect. By doing this, multi-head attention ensures that the input embeddings are updated from a more varied and diverse set of perspectives. After the attention outputs from all heads are calculated, they are concatenated and passed through a final linear transformation to generate the output.

Section 4

$$P E (p o s , 2 i) = \sin \left(\frac{p o s}{10000^{2i/d_{model}}} \right)$$

Section 5

AI boom era As early as spring 2017, even before the "Attention is all you need" preprint was published, one of the co-authors applied the "decoder-only" variation of the architecture to generate fictitious Wikipedia articles. Transformer architecture is now used alongside many generative models that contribute to the ongoing AI boom. The "reference implementation" of the original Transformer was written in a TensorFlow library. In language modelling, ELMo (2018) was a bi-directional LSTM that produces contextualized word embeddings, improving upon the line of research from bag of words and word2vec. It was followed by BERT (2018), an encoder-only transformer model. In October 2019, Google started using BERT to process search queries. In 2020, Google Translate replaced the previous RNN-encoder–RNN-decoder model by a transformer-encoder–RNN-decoder model. Starting in 2018, the OpenAI GPT series of decoder-only transformers became state of the art in natural language generation. At the end of 2022, ChatGPT, a chatbot based on a fine-tuned variant of GPT-3.5, became unexpectedly popular, triggering a boom around large language models. Transformers have been applied in modalities beyond text. Four days after the publication of "Attention is All You Need", a multimodal transformer architecture, MultiModel, was published by most authors of that paper. Other examples include the vision transformer, speech recognition, robotics, and multimodal. The vision transformer, in turn, stimulated new developments in convolutional neural networks. Image and video generators like DALL-E (2021), Stable Diffusion 3 (2024), and Sora (2024), use transformers to analyse input data (like text prompts) by breaking it down into "tokens" and then calculating the relevance between each token using self-attention, which helps the model understand the context and relationships within the data.

Section 6

Positional encoding Since the Transformer does not rely on recurrence or convolution of the text in order to perform encoding and decoding, the paper relied on the use of sine and cosine wave functions to encode the position of the token into the embedding. The methods introduced in the paper are discussed below:

Section 7

Dataset - The English-to-German translation model was trained on the 2014 WMT (Workshop on Statistical Machine Translation) English-German dataset, consisting of nearly 4.5 million sentences derived from TED Talks and high-quality news articles. A separate translation model was trained on the much larger 2014 WMT English-French dataset, consisting of 36 million sentences. Both datasets were encoded with byte-pair encoding. **Hardware** - The models were trained using 8 NVIDIA P100 GPUs. The base models were trained for 100,000 steps, and the big models were trained for 300,000 steps - each step taking about 0.4 seconds to complete for the base models and 1.0 seconds for the big models. The base model was trained for a total of 12 hours, and the big model was trained for a total of 3.5 days. Both the base and big models outperform the 2017 state-of-the-art in both English-German and English-French, while achieving the comparatively lowest training cost. **Hyperparameters and regularization** - For their 100M-parameter Transformer model, the authors increased the learning rate linearly for the first 4000 (warmup) steps and decreased it proportionally to the inverse square root of the current step number. Dropout layers were applied to the output of each sub-layer before normalization, the sums of the embeddings, and the positional encodings. The dropout rate was set to 0.1. Label smoothing was applied with a value of 0.1, which "improves accuracy and BLEU score".